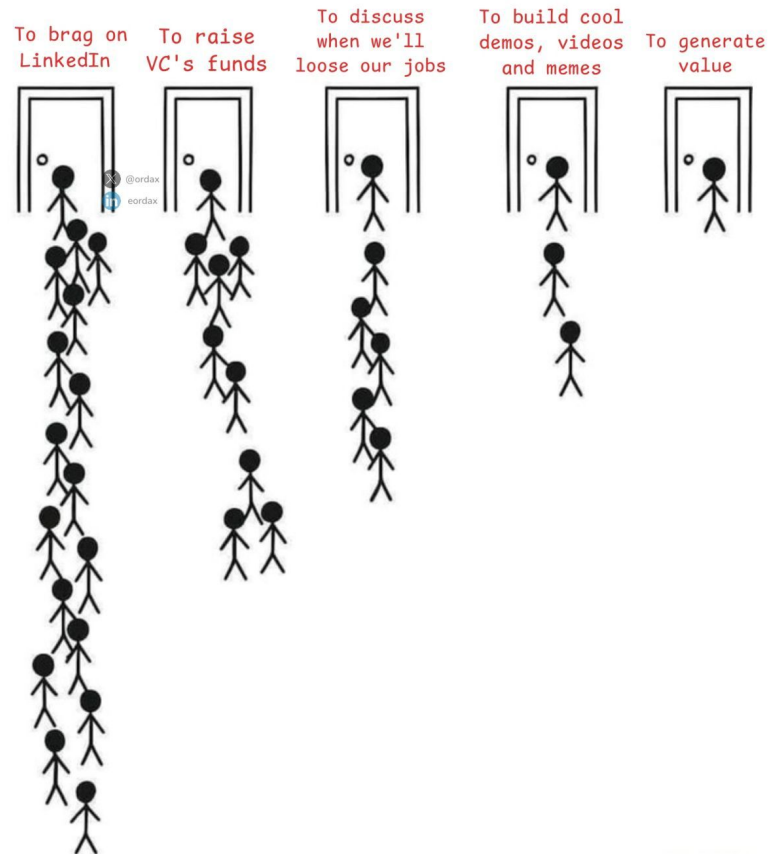


How are people using Agentic AI?

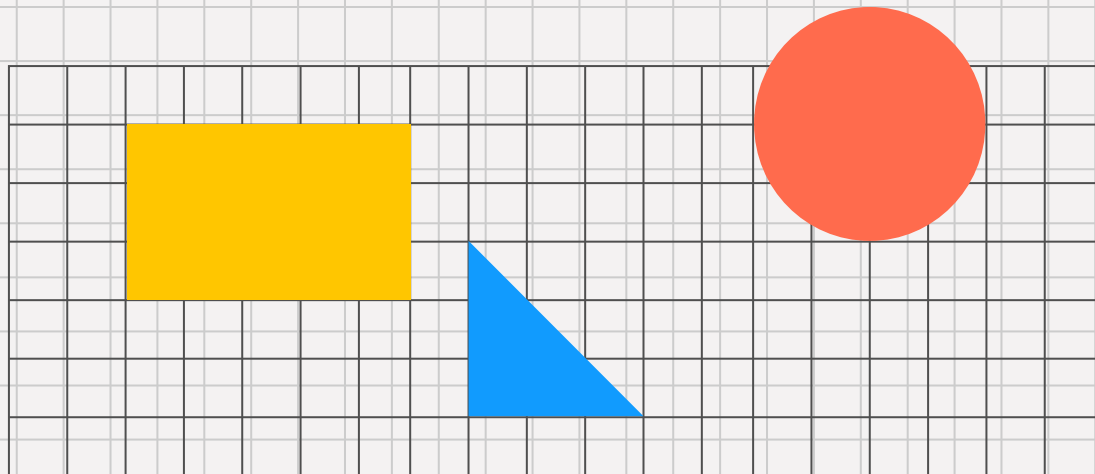


Spring 2026

IS 365

Lecture 8:

Introduction to BPM with
Agentic AI



Announcements



NEWS

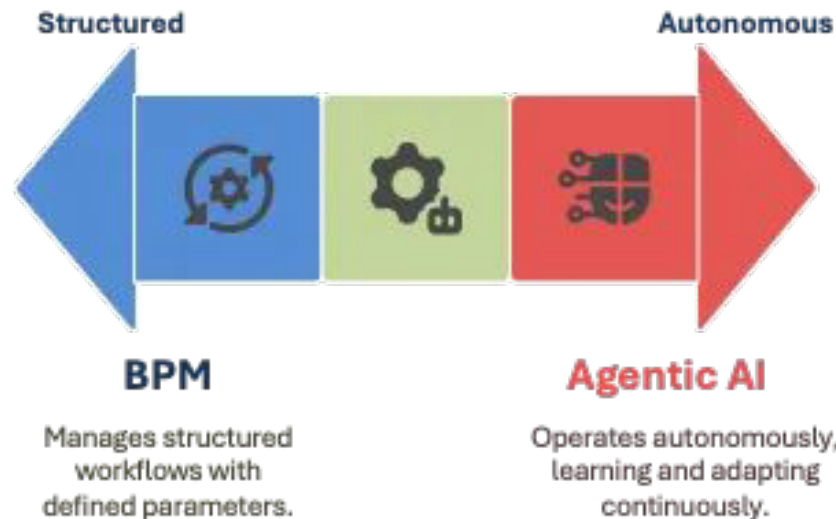
- Project Peer Feedback Survey open until Friday midnight
- Automation Workshop next week
 - Demo on Monday
 - Work on Exercise 3 as “project group” in class on Wednesday

“Future-ready enterprises will blend deterministic BPM with autonomous agents that learn and adapt. These hybrid approaches will define enterprise agility.”

Composable Enterprise Architectures for 2025, Gartner

BPM with Agentic AI

Integrates AI for enhanced orchestration and adaptability.



What Is BPM?

- **Definition:** Orchestration of structured workflows with defined actors and SLAs
- Best for structured, repeatable, regulated processes
- **Strengths:** Predictability, auditing, compliance
- **Limitations:** Rigid to change, limited contextual awareness

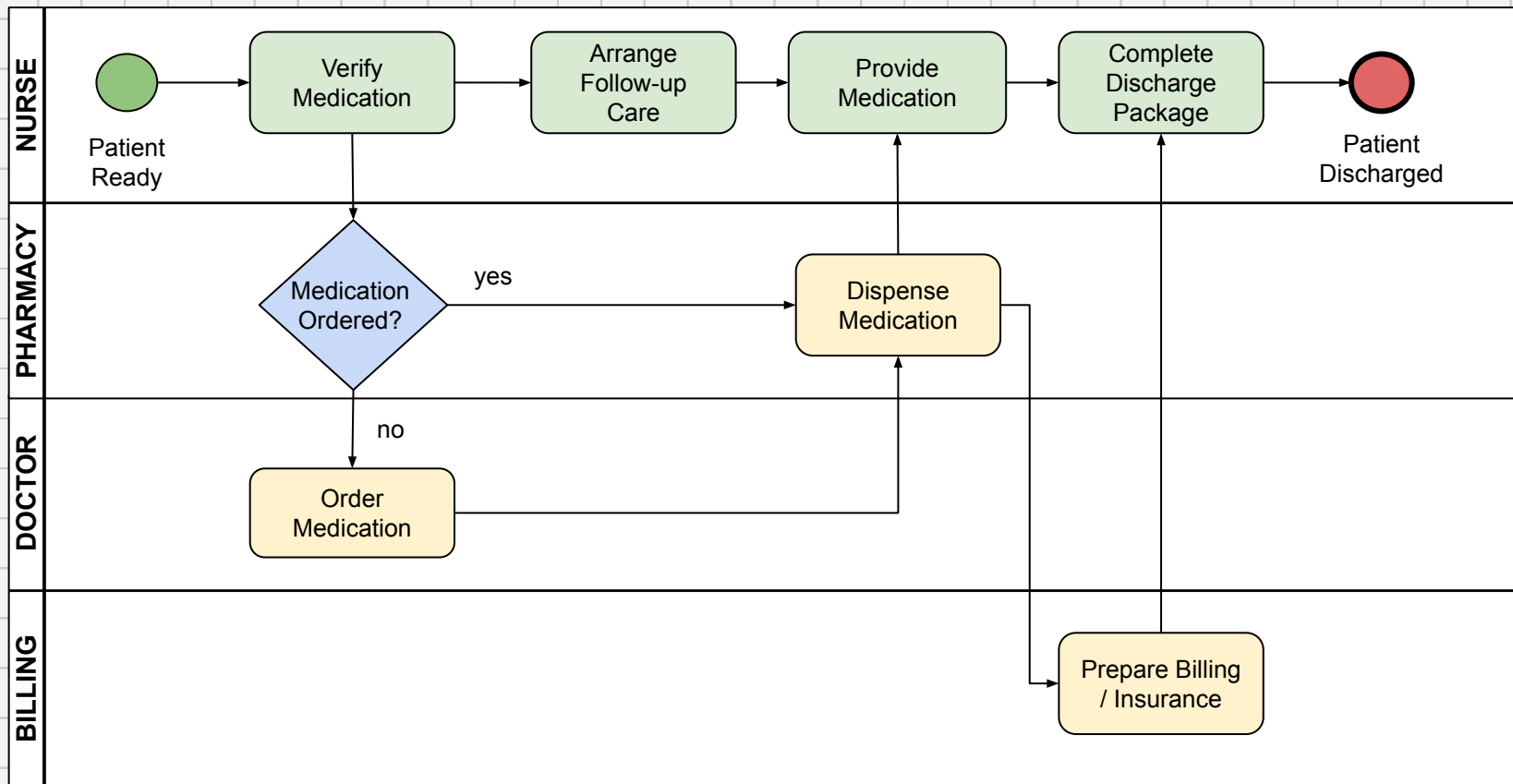
What Is Agentic AI?

- Autonomous AI 'agents' that perceive, reason, and act to achieve goals
- Best for **dynamic, context-driven processes**
- **Strengths:** Adaptability, contextual reasoning, autonomy
- **Limitations:** Less predictable, needs governance

Hybrid — The Middle Ground

- Combines structured BPM with autonomous AI
- Governance from BPM + adaptability from AI
- Ideal for workflows with structured core and dynamic edge cases
- Reduces costs and improves turnaround times

Case Study: Hospital Discharge Process



Hospital Discharge Process

- **Involves multiple departments:** doctors, pharmacy, billing, insurance, nurses
- **Goal:** faster discharge, fewer handoffs, higher satisfaction
- BPM defines process; AI adds adaptability

BPM vs Agentic AI Roles

- BPM orchestrates task order, tracking, and notifications
- Agentic AI resolves insurance queries and suggests medication alternatives
- AI generates summaries and patient communication

AI Agents

- “Discharge Coordination” Agent
- “Insurance Validation” Agent
- “Pharmacy Fulfillment” Agent
- “Billing” Agent
- “Communication” Agent

Results

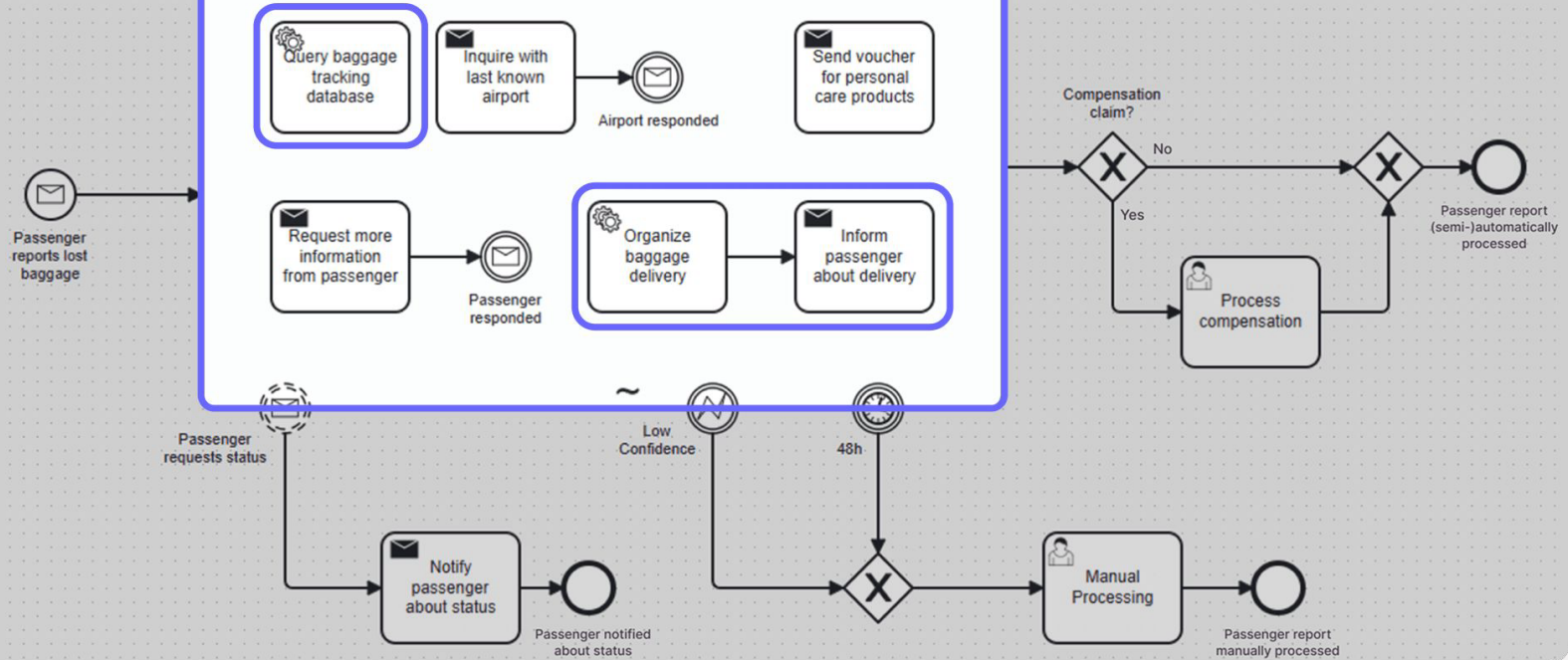
- 20–30% productivity gain
- Discharge time reduced from 5 hrs → <1 hr
- SLA compliance ~98%, patient satisfaction +15–20%

Another Example

AI Agent

The BPMN ad-hoc subprocess leverages any LLM to decide the next best action

Baggage Loss Agent



When to Use What

- **BPM-only:** stable, regulated processes
- **Agentic AI-only:** dynamic, exploratory, creative processes
- **Hybrid:** structured governance + adaptive intelligence

Approach	When to Use
BPM	• Structured, rule-based flows with predictable actors and decisions • Heavily regulated, repeatable workflows
Agentic AI	• Unstructured, dynamic tasks needing perception, language, or goal-seeking AI • Dynamic, data-rich, outcome-driven use cases
Hybrid	• Semi-structured workflows with variability, data gaps, or automation gaps • Core processes with dynamic decision edges

Capability	BPM Only	Agentic AI Only	Hybrid (BPM + Agentic AI)
Structure & Auditability	✅ High	⚠️ Low	✅ High
Adaptability to Edge Cases	⚠️ Low	✅ High	✅ High
Compliance & Governance	✅ High	⚠️ Medium	✅ High
Task Autonomy & Reasoning	❌ None	✅ Full	✅ Partial
Time to Value	⚠️ Medium	⚠️ Medium	✅ High (when modularized)
Learning & Optimization	❌ None	✅ Continuous	✅ Contextual Learning

Human-in-the-Loop Integration

- Keep humans for oversight and key approvals
- AI handles repetitive or analytical subtasks
- Collaboration improves both speed and accountability

Governance and Explainability

- BPMN provides transparent process trail
- AI adds reasoning transparency and monitoring
- **Together:** accountable automation with audit history

Workshop

Guest Speaker



Automotive Racing
International Travel

Hobbies

Arjun Chauhan

Current: Product Manager, Bravo Store Systems

Experience: Process Optimization, Power BI Reporting, Azure Data Factory Architecture, Cybersecurity Workflows

Expertise: Agentic AI, Data Systems, Supply Chain Modeling, AI-driven Knowledge Retrieval